

STS 531: Mathematical Statistics I

Homework IV

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4.4 A pdf is defined by

$$f(x, y) = \begin{cases} C(x + 2y) & \text{if } 0 < y < 1 \text{ and } 0 < x < 2 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find the value of C .
- (b) Find the marginal distribution of X .
- (c) Find the joint cdf of X and Y .
- (d) Find the pdf of the random variable $Z = 9/(X + 1)^2$.

Solution: (a) For f to be a valid pdf, we must have

$$1 = \int_0^1 \int_0^2 C(x + 2y) \, dx dy \tag{1}$$

$$= C \int_0^1 2 + 4y \, dy \tag{2}$$

$$= 4C. \tag{3}$$

Therefore, $C = \frac{1}{4}$.

(b) The pdf of the marginal distribution of X is

$$f_X(x) = \int_0^1 \frac{1}{4}(x + 2y) \, dy \tag{4}$$

$$= \frac{1}{4} [xy + y^2]_0^1 \tag{5}$$

$$= \frac{1}{4}(x + 1), \tag{6}$$

for $0 < x < 2$.

(c) The joint cdf of X and Y is

$$F_{X,Y}(x, y) = \int_0^y \int_0^x \frac{1}{4}(u + 2v) \, dudv \quad (7)$$

$$= \frac{1}{4} \int_0^y \left[\frac{x^2}{2} + 2xv \right] dv \quad (8)$$

$$= \frac{1}{4} \left(\frac{x^2 y}{2} + xy^2 \right) \quad (9)$$

$$= \frac{x^2 y}{8} + \frac{xy^2}{4}. \quad (10)$$

(d) If $Z = g(X) = 9/(X + 1)^2$, then $g^{-1}(Z) = \frac{3}{\sqrt{Z}} - 1$. Using Theorem 2.1.5 of the textbook (pg. 51), the pdf of Z is

$$f_Z(z) = f_X(g^{-1}(z)) \left| \frac{d}{dz} g^{-1}(z) \right| \quad (11)$$

$$= \frac{1}{4} \left(\frac{3}{\sqrt{z}} \right) \left(\frac{3}{2z^{\frac{3}{2}}} \right) \quad (12)$$

$$= \frac{9}{8} z^{-2}, \quad (13)$$

for $1 < z < 9$. Outside of this range, $f_Z(z)$ is 0.

4.10 The random pair (X, Y) has the distribution

		X		
		1	2	3
Y	2	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{12}$
	3	$\frac{1}{6}$	0	$\frac{1}{6}$
	4	0	$\frac{1}{3}$	0

(a) Show that X and Y are dependent.

(b) Give a probability table for random variables U and V that have the same marginals as X and Y but are independent.

Solution: (a) Note that $P(X = 1) = \frac{1}{12} + \frac{1}{6} = \frac{1}{4}$ and $P(Y = 4) = \frac{1}{3}$. Therefore, $P(X = 1)P(Y = 4) = \frac{1}{12}$. However, $P(X = 1, Y = 4) = 0$. Thus, X and Y are dependent.

(b) U and V will be independent and have the same marginals as X and Y if their probability table is

		U		
		1	2	3
V	2	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{12}$
	3	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{12}$
	4	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{12}$

4.15 Let $X \sim \text{Poisson}(\theta)$, $Y \sim \text{Poisson}(\lambda)$ be independent random variables. It was shown in Theorem 4.3.2 that the distribution of $X + Y$ is $\text{Poisson}(\theta + \lambda)$. Show that the distribution of $X|X + Y$ is binomial with success probability $\theta/(\theta + \lambda)$. What is the distribution of $Y|X + Y$?

Solution: Let $U = X + Y$ and $V = X$. Then the pdf of $X|X + Y$, which is the same as the pdf of $V|U$, is

$$f_{V|U}(v|u) = \frac{f_{U,V}(u, v)}{f_U(u)}. \quad (14)$$

From Example 4.3.1 from the textbook, we know that

$$f_{U,V}(u, v) = \frac{\lambda^{u-v} e^{-\lambda}}{(u-v)!} \frac{\theta^v e^{-\theta}}{v!}$$

and

$$f_U(u) = \frac{e^{-(\theta+\lambda)}}{u!} (\theta + \lambda)^u.$$

Therefore,

$$f_{V|U}(v|u) = \frac{f_{U,V}(u, v)}{f_U(u)} \quad (15)$$

$$= \left(\frac{u!}{(u-v)!v!} \right) \left(\frac{\lambda^{u-v}\theta^v}{(\theta+\lambda)^u} \right) \quad (16)$$

$$= \binom{u}{v} \left(\frac{\lambda}{\theta+\lambda} \right)^{u-v} \left(\frac{\lambda}{\theta+\lambda} \right)^v \left(\frac{\theta}{\lambda} \right)^v \quad (17)$$

$$= \binom{u}{v} \left(\frac{\lambda}{\theta+\lambda} \right)^{u-v} \left(\frac{\theta}{\theta+\lambda} \right)^v. \quad (18)$$

This is the pdf of a binomial random variable with u trials and probability of success $\frac{\theta}{\theta+\lambda}$.

The distribution of $Y|X + Y$ will be binomial with u trials and probability of success $\frac{\lambda}{\theta+\lambda}$.

4.40 A generalization of the beta distribution is the Dirichlet distribution. In its bivariate version, (X, Y) have pdf

$$f(x, y) = Cx^{a-1}y^{b-1}(1-x-y)^{c-1}, \quad 0 < x < 1, \quad 0 < y < 1, \quad 0 < y < 1-x < 1,$$

where $a > 0$, $b > 0$, and $c > 0$ are constants.

(a) Show that $C = \frac{\Gamma(a+b+c)}{\Gamma(a)\Gamma(b)\Gamma(c)}$.

(b) Show that, marginally, both X and Y are beta

(c) Find the conditional distribution of $Y|X = x$, and show that $Y/(1-x)$ is $\text{beta}(b, c)$.

(d) Show that $E(XY) = \frac{ab}{(a+b+c+1)(a+b+c)}$, and find their covariance.

Solution: (a) For f to be a valid pdf, we must have

$$1 = C \int_0^1 \int_0^{1-y} x^{a-1} y^{b-1} (1-x-y)^{c-1} dx dy. \quad (19)$$

Let $t = \frac{x}{1-y}$. Then $dt = \frac{1}{1-y} dx$, so $dx = (1-y)dt$. We now have

$$1 = C \int_0^1 \int_0^1 [t(1-y)]^{a-1} y^{b-1} (1-t(1-y)-y)^{c-1} (1-y) dt dy \quad (20)$$

$$= C \int_0^1 \int_0^1 t^{a-1} (1-y)^{a-1} y^{b-1} (1-t-ty-y)^{c-1} (1-y) dt dy \quad (21)$$

$$= C \int_0^1 \int_0^1 t^{a-1} (1-y)^{a-1} y^{b-1} [(1-y)(1-t)]^{c-1} (1-y) dt dy \quad (22)$$

$$= C \int_0^1 \left[\int_0^1 t^{a-1} (1-t)^{c-1} dt \right] y^{b-1} (1-y)^{a+c-1} dy \quad (23)$$

$$= C \beta(a, c) \beta(b, a+c). \quad (24)$$

Therefore,

$$C = \frac{1}{\beta(a, c) \beta(b, a+c)} = \frac{\Gamma(a+b+c)\Gamma(a+c)}{\Gamma(a+c)\Gamma(a)\Gamma(b)\Gamma(c)} = \frac{\Gamma(a+b+c)}{\Gamma(a)\Gamma(b)\Gamma(c)} \quad (25)$$

(b) The marginal distribution of X is

$$f_X(x) = \frac{\Gamma(a+b+c)}{\Gamma(a)\Gamma(b)\Gamma(c)} \int_0^{1-x} x^{a-1} y^{b-1} (1-x-y)^{c-1} dy. \quad (26)$$

Let $t = \frac{y}{1-x}$. Then $dt = \frac{dy}{1-x}$, so $dy = (1-x)dt$. Using this substitution, we have

$$f_X(x) = \frac{\Gamma(a+b+c)}{\Gamma(a)\Gamma(b)\Gamma(c)} \int_0^1 x^{a-1} t^{b-1} (1-x)^{b-1} (1-x-t(1-x))^{c-1} (1-x) dt \quad (27)$$

$$= \frac{\Gamma(a+b+c)}{\Gamma(a)\Gamma(b)\Gamma(c)} \int_0^1 x^{a-1} t^{b-1} (1-x)^b (1-x)^{c-1} (1-t)^{c-1} dt \quad (28)$$

$$= \frac{\Gamma(a+b+c)}{\Gamma(a)\Gamma(b)\Gamma(c)} \left[\int_0^1 t^{b-1} (1-t)^{c-1} dt \right] x^{a-1} (1-x)^{b+c-1} \quad (29)$$

$$= \left(\frac{\Gamma(a+b+c)}{\Gamma(a)\Gamma(b)\Gamma(c)} \right) \left(\frac{\Gamma(b)\Gamma(c)}{\Gamma(b+c)} \right) x^{a-1} (1-x)^{b+c-1} \quad (30)$$

$$= \frac{\Gamma(a+b+c)}{\Gamma(a)\Gamma(b+c)} x^{a-1} (1-x)^{b+c-1} \quad (31)$$

$$= \beta(a, b+c) x^{a-1} (1-x)^{b+c-1} \quad (32)$$

for $0 < x < 1$. This is a Beta Distribution with parameters a and $b+c$. By symmetry, the marginal distribution of Y will be Beta with parameters b and $a+c$.

(c) The conditional distribution of $Y|X=x$ has pdf

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)} \quad (33)$$

$$= \frac{\Gamma(b+c)}{\Gamma(b)\Gamma(c)} \frac{x^{a-1} y^{b-1} (1-x-y)^{c-1}}{x^{a-1} (1-x)^{b+c-1}} \quad (34)$$

$$= \frac{\Gamma(b+c)}{\Gamma(b)\Gamma(c)} \frac{y^{b-1} (1-x-y)^{c-1}}{(1-x)^{b+c-1}}. \quad (35)$$

Let $Z = g(Y) = \frac{Y}{1-x}$. Then $g^{-1}(Z) = Z(1-x)$. Using Theorem 2.1.5, the pdf of Z is

$$f_Z(z) = f_{Y|X}(g^{-1}(z)) \left| \frac{d}{dz} g^{-1}(z) \right| \quad (36)$$

$$= \frac{\Gamma(b+c)}{\Gamma(b)\Gamma(c)} \frac{[z(1-x)]^{b-1} (1-x-z(1-x))^{c-1}}{(1-x)^{b+c-1}} (1-x) \quad (37)$$

$$= \frac{\Gamma(b+c)}{\Gamma(b)\Gamma(c)} \frac{z^{b-1} (1-x)^b ([1-x][1-z])^{c-1}}{(1-x)^{b+c-1}} \quad (38)$$

$$= \frac{\Gamma(b+c)}{\Gamma(b)\Gamma(c)} \frac{z^{b-1} (1-x)^{b+c-1} (1-z)^{c-1}}{(1-x)^{b+c-1}} \quad (39)$$

$$= \frac{\Gamma(b+c)}{\Gamma(b)\Gamma(c)} z^{b-1} (1-z)^{c-1} \quad (40)$$

for $0 < z < 1$. Thus, $Y/(1-x)$ is beta(b, c).

(d) We can use the result of part (a) to solve this. The expected value of XY is

$$E[XY] = C \int_0^1 \int_0^{1-y} x^{(a+1)-1} y^{(b+1)-1} (1-x-y)^{c-1} dx dy \quad (41)$$

$$= C \beta(a+1, c) \beta(b+1, a+1+c) \quad (42)$$

$$= \frac{\Gamma(a+b+c)}{\Gamma(a)\Gamma(b)\Gamma(c)} \frac{\Gamma(a+1)\Gamma(c)\Gamma(b+1)\Gamma(a+1+c)}{\Gamma(a+1+c)\Gamma(a+b+c+2)} \quad (43)$$

$$= \frac{\Gamma(a+b+c)ab}{\Gamma(a+b+c+2)} \quad (44)$$

$$= \frac{\Gamma(a+b+c)ab}{(a+b+c+1)(a+b+c)\Gamma(a+b+c)} \quad (45)$$

$$= \frac{ab}{(a+b+c+1)(a+b+c)} \quad (46)$$

The covariance of X and Y is

$$Cov(X, Y) = E[XY] - E[X]E[Y] \quad (47)$$

$$= \frac{ab}{(a+b+c+1)(a+b+c)} - \frac{ab}{(a+b+c)(a+b+c)} \quad (48)$$

$$= \frac{ab(a+b+c) - ab(a+b+c+1)}{(a+b+c+1)(a+b+c)^2} \quad (49)$$

$$= \frac{-ab}{(a+b+c+1)(a+b+c)^2}. \quad (50)$$

4.42 Let X and Y be independent random variables with means μ_X , μ_Y , and variances σ_X^2 , σ_Y^2 . Find an expression for the correlation of XY and Y in terms of these means and variances.

Solution: The correlation of XY and Y is

$$\rho_{XY,Y} = \frac{\text{Cov}(XY, Y)}{\sigma_{XY}\sigma_Y}.$$

First, we simplify the top of this fraction. Since X and Y are independent, we have

$$\text{Cov}(XY, Y) = E[XY^2] - E[XY]E[Y] \quad (51)$$

$$= E[X]E[Y^2] - E[X]E^2[Y] \quad (52)$$

$$= E[X]\text{Var}(Y) \quad (53)$$

$$= \mu_X\sigma_Y^2. \quad (54)$$

As for σ_{XY} , we have

$$\sigma_{XY} = \sqrt{E[(XY)^2] - E^2[XY]} \quad (55)$$

$$= \sqrt{E[X^2]E[Y^2] - (\mu_X\mu_Y)^2}. \quad (56)$$

Using the variance formula, we can write

$$E[X^2] = (\sigma_X^2 + \mu_X^2).$$

Similarly for $E[Y^2]$. Therefore,

$$\sigma_{XY} = \sqrt{E[X^2]E[Y^2] - (\mu_X\mu_Y)^2} \quad (57)$$

$$= \sqrt{(\sigma_X^2 + \mu_X^2)(\sigma_Y^2 + \mu_Y^2) - (\mu_X\mu_Y)^2} \quad (58)$$

$$= \sqrt{\sigma_X^2\sigma_Y^2 + \sigma_X^2\mu_Y^2 + \sigma_Y^2\mu_X^2}. \quad (59)$$

Thus,

$$\rho_{XY,Y} = \frac{\mu_X\sigma_Y}{\sqrt{\sigma_X^2\sigma_Y^2 + \sigma_X^2\mu_Y^2 + \sigma_Y^2\mu_X^2}}.$$